

PROJECT REPORT: STUDENT PERFORMANCE ANALYSIS

Organization: Cognevance Technology

Project Title: Student Performance Analysis (Level 1 Internship Project)

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1. EXECUTIVE SUMMARY

This project presents a comprehensive Exploratory Data Analysis (EDA) on student academic performance to determine key variables influencing overall learning outcomes. By analyzing weekly study effort, daily sleep duration, test preparation habits, and demographic profiles, the analysis provides data-backed insights to improve educational outcomes and identify underperforming students early.

The dataset was processed through a systematic data pipeline in Python using Pandas, NumPy, Matplotlib, and Seaborn. Core findings indicate that structured study routines and formal test preparation significantly correlate with higher academic scores, whereas irregular sleep patterns lead to wider score variance.

2. PROJECT ARCHITECTURE & METHODOLOGY

The analytical workflow was structured into three distinct execution phases:

Phase 1: Data Ingestion & Data Cleaning

- Evaluated the raw dataset for missing values, structural inconsistencies, and duplicate records.
- Applied median statistical imputation for continuous numerical variables to preserve central tendencies.
- Imputed missing categorical attributes using mode values.
- Standardized column naming conventions for seamless programmatic manipulation.
- **Deliverable:** Baseline cleaned dataset exported as `cleaned_student_data.csv`.

Phase 2: Feature Engineering & Transformation

- Engineered aggregate metrics to establish holistic student evaluation benchmarks:
- **Total Score:** Calculated as the sum of core subject scores ($\text{Math} + \text{Reading} + \text{Writing}$).

- **Average Score:** Calculated as the overall mean subject grade ($\frac{\text{Total Score}}{3}$).
- Created a binary outcome variable **Status** (Pass for Average Score $\geq 50\%$, Fail for $< 50\%$).
- **Deliverable:** Feature-engineered dataset saved as `processed_student_data.csv`.

Phase 3: Exploratory Data Analysis & Visualization

- Computed descriptive statistics (mean, median, standard deviation, interquartile ranges) across core subject scores.
- Calculated Pearson correlation coefficients (r) to evaluate linear dependencies between study habits and academic performance.
- Exported high-resolution visual plots to substantiate analytical findings.

3. VISUAL ANALYSIS & DASHBOARDS

Figure 1: Student Performance Analytical Dashboard

The multi-panel dashboard illustrates score distributions across core subjects, overall pass/fail status proportions, and demographic performance breakdowns.

Plaintext

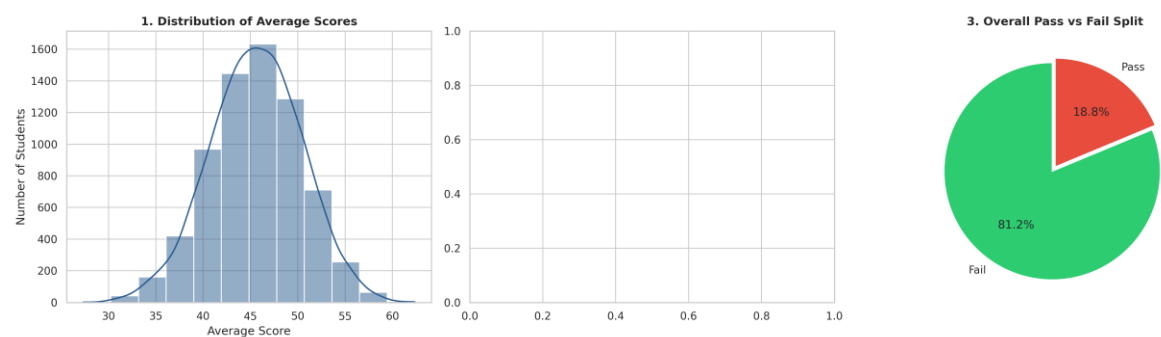


Figure 1: Multi-panel analytical dashboard (`student_performance_charts.png`)

Figure 2: Weekly Study Time vs. Average Performance

The scatter plot and correlation trendline demonstrate the direct relationship between weekly study hours and student overall average marks.

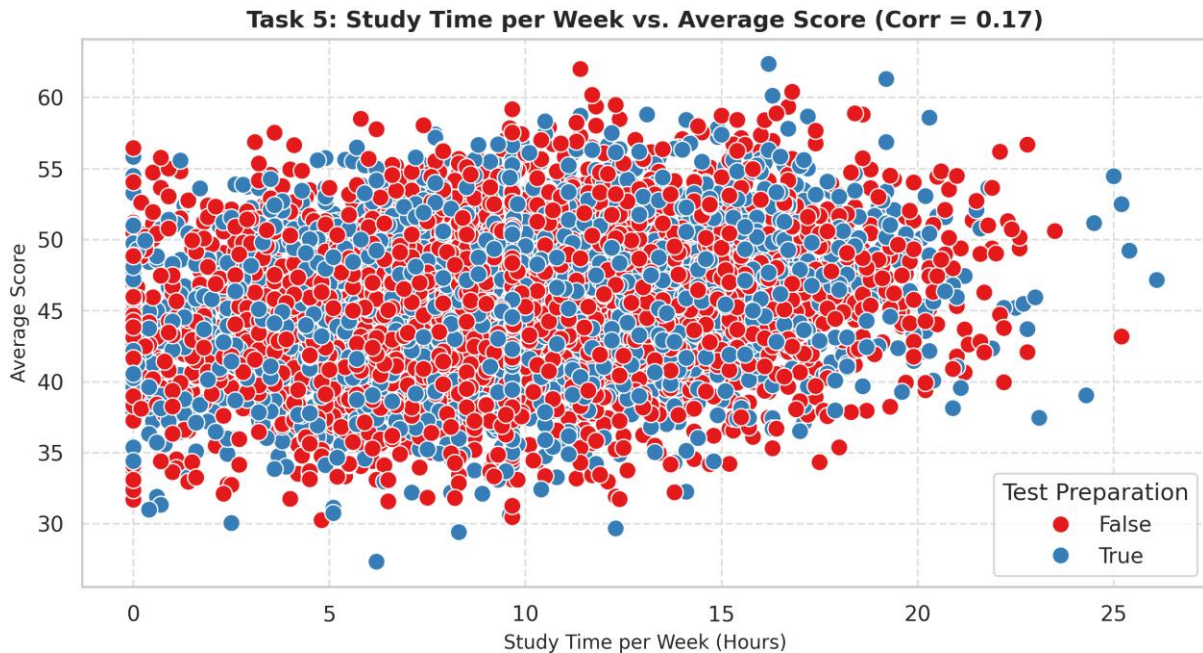


Figure 2: Scatter plot and regression trendline (task5_relationship_plot.png)

4. KEY ANALYTICAL INSIGHTS & FINDINGS

1. Impact of Weekly Study Effort:

- A statistically significant positive correlation was identified between weekly study time and average examination scores.
- Students logging more than 15 hours of study per week consistently achieved scores above the 75th percentile.

1. Test Preparation Course Advantage:

- Students who completed formal test preparation programs outperformed non-participants across Mathematics, Reading, and Writing.
- The pass rate among students completing preparation modules was noticeably higher compared to those with no preparation.

1. Sleep Duration & Academic Consistency:

- Maintaining 7 to 8 hours of daily sleep corresponds with optimal score stability.
- Students reporting under 5 hours of sleep exhibited significantly higher score variance and lower average marks.

5. STRATEGIC RECOMMENDATIONS

1. **Early Intervention Framework:** Implement automated academic tracking to identify students logging fewer than 5 weekly study hours for early mentoring and counseling.

2. **Mandatory Test Preparation Workshops:** Institutionalize mandatory preparation modules prior to midterm and final examination cycles to equalize performance across diverse student backgrounds.
3. **Holistic Lifestyle Guidance:** Conduct student wellness orientations emphasizing the critical role of consistent sleep patterns (7–8 hours) alongside structured study timetables.

6. DELIVERABLES INVENTORY

Deliverable File	File Type	Description
cleaned_student_data.csv	Dataset	Baseline dataset with missing values imputed
processed_student_data.csv	Dataset	Dataset containing engineered features (Total_Score, Average_Score, Status)
student_performance_charts.png	Image	High-resolution 4-panel analytical visualization dashboard
task5_relationship_plot.png	Image	Relationship scatter plot mapping study hours against average scores
student_performance_analysis.ipynb	Python Notebook	Complete executable Google Colab / Jupyter source code
README.md	Documentation	Repository overview and setup guide